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1/21/2025

2025 Spring Project Plan

Managing Resources

- Form subgroups that align with interests
- Analyze problem and solution groupings
- Database implementation for uploading notes
- Sentiment analysis performed on the resources
- Utilizing Figma + LLM to tag and demonstrate the functionality with dummy data.

Testing → System Prompts / User-Flow / Survey / Algorithm

1/28/2025

Prototype development & testing

- Step 1: Identify app goal & key features to prototype
- Step 2: Sketch primary user flows & wireframes
- Step 3: Turn wireframes into prototypes
- Step 4: Demo prototype.

Prototype goal & key features

- metrics like posts shared, tag relevance
- Sentiment analysis using LLM ⇒
 - used to analyze notes & provide summary
 - Rank resources based on reliability of contributors [professor recommended / high karma]

01/28/25

Prototype testing

- visual walkthrough with Aigme
- testing with synthetic data
- input & output data

Leaderboard feature to encourage sharing of resources

Feedback: Questions to analyze before next week

Prototype key feature identification & testing

Identify target group and method to test

2/4/2025 Guest Speaker: Daniel Losagie

MVP mindset:

1. Listen (Research)

2. Design

3. Develop

4. Test



Testing data:

→ Do tests with different user groups

→ Market your development

(Screen studio)

Collecting data:

→ Required data

→ Obtained through

→ Stored in

Designing online

→ Figma / Play

→ Prototype

→ Inspiration

from companies

Testing Developing:

→ IDE

→ LLM

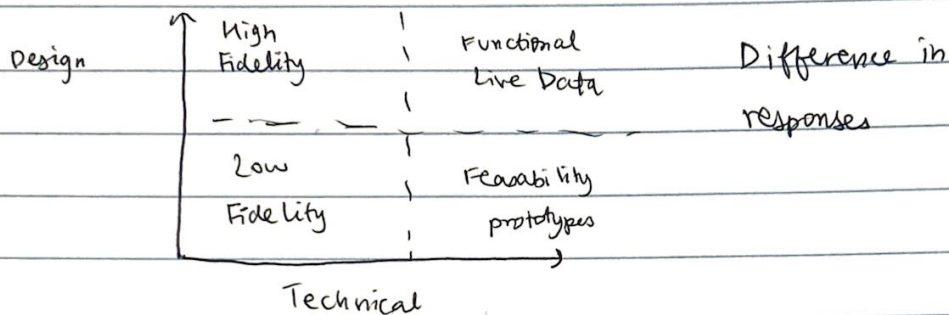
→ Frameworks for data testing

} can speed up work

Prototypes → are early models of projects

2/4/2025

Types of prototype testing :



My contributions : Sentiment analysis for resources & tagging



- Present at demo
- Planning testing phase of this

2/11/2025

Demo of prototype - to-do list

Feedback :

- ✓ Scope of AI tagging
- ✓ Testing criteria
- ✓ Accuracy of material
- ✓ Accuracy of tagging system

Prototype Test Plan - worksession

02/18/25

- Tagging based on AI based on topic list - help students.
- Use Rock & AGENTA to collaborate and test prompts
- Quality / accuracy of resources comparing to teacher's notes, consistency etc.
- Testing different prompts
- arXiv journal repository

02/25/25

Phase 2 - Testing

Where are we?

- Rubric for evaluating LLM responses to accuracy of the material
- Combine with sentiment analysis of response to benchmark and rank response effectiveness
- Prompt-engineering to determine which method finds the most accurate tags for uploaded resources, comparing it to a standard resource that has been uploaded by an instructor (eg: teacher)
- Prepare slides to discuss test-findings.
- Testing prompts for matching tags to user posts ★

Testing: Important features

- Stakeholder engagements
- Concept testing - design and plan

3/11/2025

Demo of testing plan

1. Tagging resources

Idea:

Syllabus
+

notes



Tags for resources

- Extract key topics from the notes by identifying important concepts, grouping related terms, comparing it to the syllabus.

- Map these topics to the syllabus by finding direct or partial matches with course sections, then output a structured alignment highlighting relevant overlaps.

To-do → Find appropriate prompt: [My contribution]

↳ Use Agenta.ai to test prompts through multiple iterations

Testing: manual validation

Record variations in prompts

Architecture diagrams

Maintain record of prompts & efficiency & why it was picked.

System of tagging resources → User picks tags?

or just fed in by system and resources

↓
~~delete~~ test and find

2. Sentiment Analysis of Resources

- Provide detailed rubric to LLM?
- LLM ranks resources on multiple factors on a score from 1 to 5.

03/11/2025 Prompt engineering using Agenta. → My work → to-do-checklist

- Iterations, final one had the most accurate tags → manually validated
- Design front-end UI for tagging system using react → integrate with other parts of system
- Verify the resources from CS-50, check the resources against the provided rubric
- Use Gemini API key - free to use and provides relatively fast responses

LLM for feedback system:

- Should user be aware of feedback on their resource?
- How should resources be ranked in the system?
- Would it lead to a popularity bias?

3/25/2025

Guest Speaker - Carter

Topic: Augmenting transactive memory systems for collectively intelligent teams

- Balance high performance & quality of work
- Transactive memory system → "who knows what"

↳ credibility, co-ordination

↳ reduced cognitive load, enhanced team performance

Collective intelligence - 'teams of teams' solve complex problems

AI and future of work → connecting people to fill gaps

~~ok~~ Group discussion

04/01/2025

- Split into groups for discussion and preparation

04/08/2025

for the final demo

- we worked on our slides and decided who would do what part

- Prepared working prototype of the web-app with complete functionality

- should be able to process API calls and return results in real-time.

- Current limitations & future directions discuss

- Problem Statement:

Students struggle to find & organize resources efficiently

Goal:

Build Web-App to tag posts, tag helpful contributors

4/15/2025

Final Demo : Spring 2025

Our presentation plan:

Walkthrough of demo

Tagging resources → how it works, how it looks

Purpose of it

Feedback mechanism → rubric for evaluation

feedback UI, scores and description

Current Limitations → notes quality & accuracy check

→ popularity bias could arise

Future Direction → Instructors should verify LLM scores

→ Advanced tagging with semantic analysis

→ Personalized recommendations based on student profiles.

→ Gamified design for incentive to upload.

Feedback from viewers:

- Integration with handwritten notes & other forms
- Comparison with instructor notes for accuracy alone could be flawed (may not account for what is not on slides for example)
- Integration of same color scheme as other subgroups